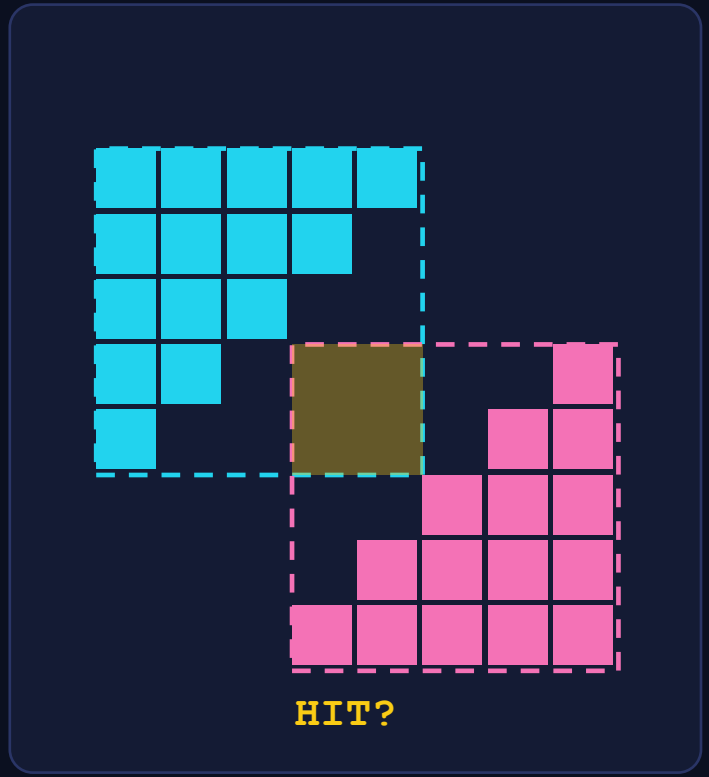
CCA GAME DEVELOPMENT

Pixel-Perfect Collisions

Module 2 — Session 8

Making Collisions More Precise

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LEVEL 01 · TODAY'S MISSION

Today's Mission

Today, you will improve collision detection in your game.
You will learn how to:

1. Understand why rectangle collisions are not always accurate.
2. Compare rectangle collisions and pixel-perfect collisions.
3. Create collision masks using `pygame.mask`.
4. Detect collisions between irregular sprite shapes.
5. Upgrade and test collision detection in an existing game.



TODAY'S GOAL

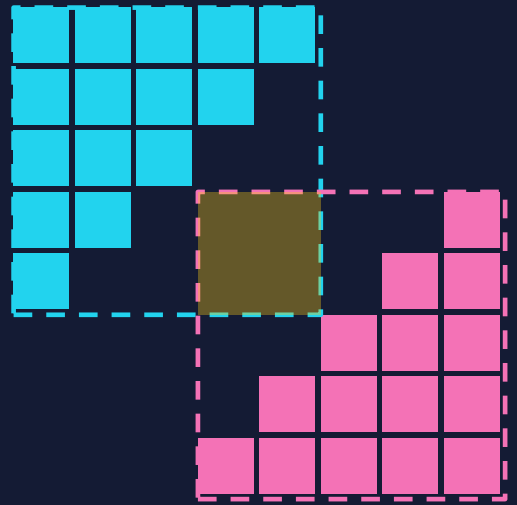
Make collision detection **more precise** by checking the **visible parts** of sprites.

When Rectangles Lie

A sprite's `Rect` **represents its rectangular boundary, not its actual shape.**

RECT COLLISION

- Checks whether two rectangles overlap.
- Fast and simple.
- May detect a collision even when visible pixels do not touch.

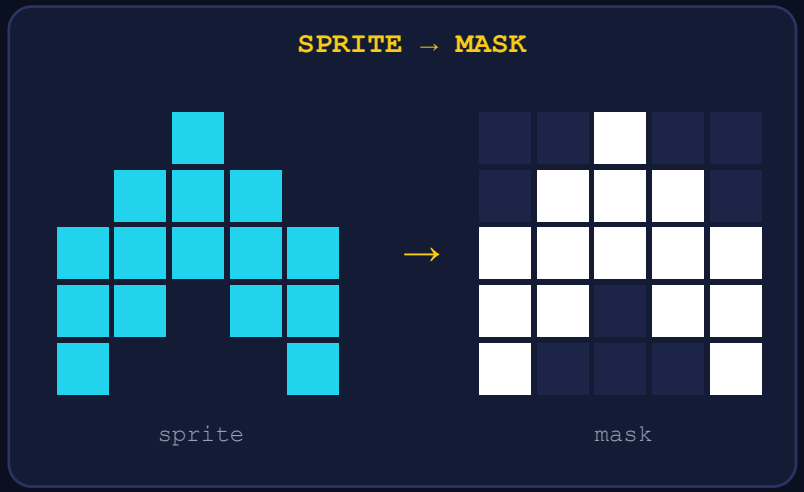


Rects overlap.
Visible pixels don't.

Meet the Collision Mask

A collision mask represents which pixels of a sprite are solid for collision detection.

- Opaque pixels are included.
- Transparent pixels are ignored.
- Two masks are checked for overlapping pixels.



Rect

Checks rectangular boundaries.

Pixel-Perfect

Checks overlapping pixels using masks.

Key Idea: **A mask detects collisions based on the actual sprite shape, instead of its entire rectangle.**

From Image to Collision

STEP 1

Create a Mask

```
self.mask = pygame.mask.from_surface(self.image)
```

Creates a mask from the sprite image's opaque pixels.

STEP 2

Check the Collision

```
pygame.sprite.collide_mask(player, enemy)
```

Checks whether the two sprites' masks overlap.

- Returns a collision point if the masks overlap.
- Returns **None** if they do not.
- Both sprites need suitable images and masks for accurate results.

Remember: **Creating the mask and checking the collision are two different steps.**

Rectangle or Pixel-Perfect?

OPTION 1

Rectangle Collision

```
if player.rect.colliderect(enemy.rect):
    print("Collision!")
```

- Checks rectangle overlap.
- Fast and simple.
- Good for rectangular objects and approximate collision detection.

OPTION 2

Pixel-Perfect Collision

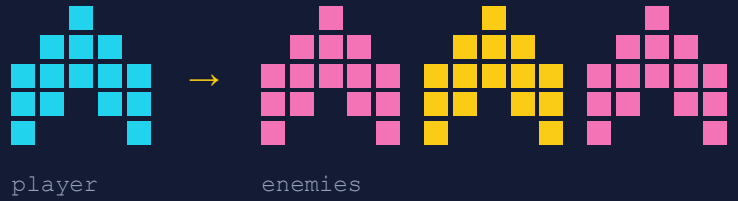
```
if pygame.sprite.collide_mask(player, enemy):
    print("Pixel Collision!")
```

- Checks mask overlap.
- More precise for irregular sprite shapes.
- Can require more processing.

Mask Collisions with Groups

```
hits = pygame.sprite.spritecollide(
    player,
    enemies,
    False,
    pygame.sprite.collide_mask
)
```

Checks the player against a group of enemies using mask-based collision detection.



Design Decision: Use pixel-perfect collision where precision matters. Not every object needs it.

PROJECT · MAKE IT PRECISE

Your Mission: Upgrade the Collisions

REQUIRED TASKS

1. Find a collision that feels inaccurate.
2. Create masks for the relevant sprites.
3. Replace the appropriate rectangle collision check with mask-based detection.
4. Test different sprite positions.
5. Compare the results before and after the change.

Use your existing game as the starting point.

WORKFLOW

BUILD

RUN

TEST

COMPARE

IMPROVE

Test and Compare

TEST THESE SITUATIONS

- Rectangles overlap, but visible pixels do not.
- Visible parts of the sprites touch.
- Sprites move to different positions.

BONUS CHALLENGE

- Compare the performance of rectangle and mask collisions.
- Decide which objects need pixel-perfect detection.

Success Criteria: **Collisions are more accurate, and the existing game mechanics still work.**

Today's Big Ideas

Rect Collision

Checks whether rectangular boundaries overlap.

Pixel-Perfect Collision

Checks whether opaque pixels overlap using masks.

KEY FUNCTIONS

`pygame.mask.from_surface()`	creates a mask.
`pygame.sprite.collide_mask()`	checks mask overlap.
`pygame.sprite.spritecollide()`	checks a sprite against a group.

REMEMBER

- Rectangles are fast for simple checks.
- Masks help with irregular shapes.
- More precision can need more processing.

Today's Goal: Don't just detect collisions. Detect them where they actually happen.